DonorHyperglycemiaStatsProject

Running Code

Hypothesis testing for relationship between donor demographics and delayed liver graft function. All tests are compared to a significance level of 0.05.

Donor Age Hypothesis Test

```
    Welch Two Sample t-test

data:  d_age by delayed_fn
t = -1.2831, df = 234.05, p-value = 0.2007
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 -5.648353  1.192857
sample estimates:
mean in group 0 mean in group 1
       42.10329        44.33103
```

The null hypothesis is that the mean ages of both groups are equal, while the alternative hypothesis is that the mean ages are not equal. The test statistic follows a t distribution with 234.05 degrees of freedom. The p-value is 0.2007, which is above the significance level of 0.05. Therefore, we fail to reject the null hypothesis of the mean donor ages being the same, regardless of whether or not there was delayed liver graft function.

Sex Hypothesis Test:

The null hypothesis is that there is no relationship between sex and Delayed Liver Graft Function. The alternative hypothesis is that there is a relationship.

```
    Pearson's Chi-squared test with Yates' continuity correction

data:  table(DonorHyperglycemia$d_genderf, DonorHyperglycemia$delayed_fn)
X-squared = 0.47116, df = 1, p-value = 0.4925
```

The test statistic 0.47116 follows a Chi-squared distribution with one degree of freedom. The p-value of 0.5172 is greater than the significance level of 0.4925, so we fail to reject the null hypothesis that there is no relationship. There is insufficient evidence to support there being a relationship between donor gender and delayed liver graft function.

Cause of Death Hypothesis Test

The null hypothesis is that there is no relationship between donor cause of death and delayed liver graft function. The alternative hypothesis is that there is a relationship.

```
      0   1
  1 179  72
  2 152  43
  3  84  22
  4  11   8
```

```
    Fisher's Exact Test for Count Data

data:  table(DonorHyperglycemia$d_cod, DonorHyperglycemia$delayed_fn)
p-value = 0.08737
alternative hypothesis: two.sided
```

The p-value under a fisher's exact test is 0.08737 which is greater than the significance level of 0.05, therefore we fail to reject the null. There is insufficient evidence to support a relationship between donor cause of death and delayed liver graft function

Race Hypothesis Test

The null hypothesis is that there is no relationship between donor whiteness and delayed liver graft function. The alternative hypothesis is that there is a relationship.

```
	Pearson's Chi-squared test with Yates' continuity correction

data:  table(DonorHyperglycemia$d_caucasian, DonorHyperglycemia$delayed_fn)
X-squared = 0.0067798, df = 1, p-value = 0.9344
```

The test statistic follows a chi-squared distribution with one degree of freedom. The p-value of 0.9344 is greater than the significance level of 0.5, therefore we fail to reject the null hypothesis. There is insufficient evidence to support a relationship between donor whitenesss and delayed liver graft function.

Recipient MELD Scores (Assigned and Calculated)

```r
t.test(r_meld_calc ~ delayed_fn, data = DonorHyperglycemia)
```

```
	Welch Two Sample t-test

data:  r_meld_calc by delayed_fn
t = -1.3365, df = 233.1, p-value = 0.1827
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 -2.8198807  0.5403924
sample estimates:
mean in group 0 mean in group 1 
       18.00711        19.14685 
```

```r
t.test(r_meld_assign ~ delayed_fn, data = DonorHyperglycemia)
```

```
	Welch Two Sample t-test

data:  r_meld_assign by delayed_fn
t = 0.3125, df = 241.92, p-value = 0.7549
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
95 percent confidence interval:
 -0.8728208  1.2019739
sample estimates:
mean in group 0 mean in group 1 
       22.33976        22.17518 
```

High Risk vs. Low Risk

```
chisq.test(table(DonorHyperglycemia$high_risk, DonorHyperglycemia$delayed_fn))
```

```

    Pearson's Chi-squared test with Yates' continuity correction

data:  table(DonorHyperglycemia$high_risk, DonorHyperglycemia$delayed_fn)
X-squared = 4.6205, df = 1, p-value = 0.03159
```

HEMO INSTABILITY

```
table(DonorHyperglycemia$d_genderf, DonorHyperglycemia$hemo_instability)
```

```

      0   1
  0  59 280
  1  37 175
```

```
test_result <- chisq.test(table(DonorHyperglycemia$hemo_instability, DonorHyperglycemia$d_ger
print(test_result)
```

```

    Pearson's Chi-squared test with Yates' continuity correction

data:  table(DonorHyperglycemia$hemo_instability, DonorHyperglycemia$d_genderf)
X-squared = 5.1073e-30, df = 1, p-value = 1
```

```
table(DonorHyperglycemia$high_risk, DonorHyperglycemia$hemo_instability)
```

```

        0   1
  no   57 292
  yes  12  62
```

```r
test_result <- chisq.test(table(DonorHyperglycemia$hemo_instability, DonorHyperglycemia$high_
print(test_result)
```

```
    Pearson's Chi-squared test with Yates' continuity correction

data:  table(DonorHyperglycemia$hemo_instability, DonorHyperglycemia$high_risk)
X-squared = 3.0272e-30, df = 1, p-value = 1
```

Logistic Regression

```r
m1 <- glm(delayed_fn ~ d_age + d_cod + d_caucasian + d_genderf + donorrisk + sodium,
          data = DonorHyperglycemia,
          family = "binomial")

summary(m1)
```

```
Call:
glm(formula = delayed_fn ~ d_age + d_cod + d_caucasian + d_genderf +
    donorrisk + sodium, family = "binomial", data = DonorHyperglycemia)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -5.426060   2.152075  -2.521  0.01169 *
d_age       -0.002484   0.008070  -0.308  0.75822
d_cod        0.016885   0.138697   0.122  0.90311
d_caucasian -0.060812   0.273172  -0.223  0.82384
d_genderf   -0.060390   0.244290  -0.247  0.80475
donorrisk    1.069644   0.399774   2.676  0.00746 **
sodium       0.020666   0.013984   1.478  0.13945
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 497.37  on 429  degrees of freedom
Residual deviance: 484.48  on 423  degrees of freedom
  (142 observations deleted due to missingness)
```

```
AIC: 498.48

Number of Fisher Scoring iterations: 4
```